

Lecture 2. Psychological Data

Types of study

Experimental vs. Observational

Experimental research: you *manipulate* the independent (predictor) variable and record any changes in dependent variable.

- Example: giving people different amounts of alcohol before asking them to do a driving test in a simulator.

Observational research: you simply *record* both the dependent and independent variables (no manipulation involved)

- Example: measuring the amount of alcohol consumed during pregnancy and the birth weight of the child
- It is less easy to determine cause and effect in observational studies

Within-subjects vs. Between-subjects

Within-subjects designs: all participants receive all levels of the independent variable.

- Example: participants come in on 3 days and drink a whole pint on one day, half a pint on the next and no alcohol on the final day

Between-subjects designs: participants are separated into groups; each group receive only one level of the independent variable.

- Example: participants are either given a full pint, a half pint or no alcohol.

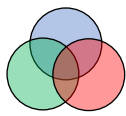
In **mixed designs** some independent variables are within-subjects and some are between.

Types of data

Name	Student (1=yes, 0=no)	Number of pets	Age (Years)
Allen	1	0	21.83
Jane	0	3	42.40
Ed	1	1	28.93
Alice	1	0	19.57
Peter	0	1	26.62
Alex	1	1	23.79
Jason	1	0	34.87

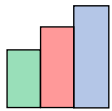
Character string	Binary (1 or 0)	Integer (whole numbers)	Real numbers (with fractions)
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Levels of measurement (N.O.I.R.)



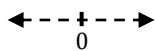
Nominal

- Categories with no logical order, e.g., breakfast cereal, mode of transport
- Can compare equality (=)



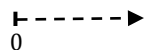
Ordinal

- Categories have an order but distance isn't meaningful, e.g., highest qualification
- Compare equality, greater than/less than (=, >/<)



Interval

- Numbers represent values at regular intervals but with no absolute zero, e.g., IQ scores/ year of birth
- Compare equality, greater than/less than and arithmetic (=, >/<, +/-)



Ratio

- Numbers with a *meaningful* zero point. 0 = no quantity of that variable, e.g., reaction time, number of siblings
- Equality, greater than/less than, arithmetic, and multiply/divide (=, >/<, +/-, x/÷)

Likert scales

Many constructs, especially ones trying to capture beliefs, attitudes, and experiences, are measured using rating scales called *Likert scales* (e.g., “On a scale from 1 to 7, how often do you...”, where responses are ordered from “Never” and “Rarely to “Often” and “All the time”).

Technically this is an *ordinal* measure, but it is typical to treat it as *interval* in practice – for example, it is common to add up responses to create a sum score or calculate a mean.

How satisfied are you with your life in general?

- ☐ Highly dissatisfied
- ☐ Somewhat dissatisfied
- ☐ Neither satisfied nor dissatisfied
- ☐ Somewhat satisfied
- ☐ Highly satisfied

Missing data

Name	Student (1=yes, 0=no)	Number of pets	Age (Years)
Allen	1	0	21.83
Jane	0	3	42.40
Ed	1	NA	28.93
Alice	1	0	19.57
Peter	0	1	26.62
Alex	1	1	23.79
Jason	1	0	34.87

Missing data points
are given the **NA**
value.

Reliability and validity

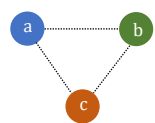
Reliability is the consistency of our measurement



- **Test-retest** reliability: consistency across time (if we repeat the measure at a later date, will we get the same result?)



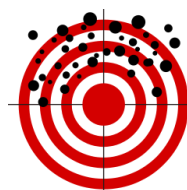
- **Inter-rater** reliability: consistency between raters (if different people administer this measure, will they get the same result?)



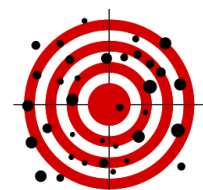
- **Internal consistency** is consistency across items (e.g., questions) in the same measure. If the measure consists of multiple questions, they should all yield similar answers. Discrepancies between items could indicate that your measure is accidentally capturing two different constructs simultaneously.

Validity is whether the measure is a good measure of the underlying construct

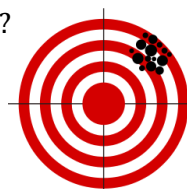
- **Face validity**: does the scale look like it measures what it's supposed to? Do others agree that it does?
- **Convergent validity**: is the test related to other tests that measure the same concept? I.e., do different measures of the construct converge on similar results?
- **Discriminant validity**: is the test capable of distinguishing between the construct of interest and an unrelated construct? E.g., a measure of reading ability should not be measuring attention.
- **Predictive validity**: is the measure related to later outcomes?



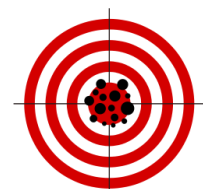
Unreliable & Invalid



Unreliable, But Valid



Reliable, Not Valid



Both Reliable & Valid

Why are reliability and validity important?

No measurement is perfect – we can estimate the degree of **measurement error** in our assessment/experimental tools.

Measurement error influences the conclusions that can be drawn:

If a measure has poor reliability, we will be unable to observe real relationships between the construct it aims to measure and another construct.

This is because the relationship between two variables can't be any stronger than the relationship between either of the variables and itself (i.e., its reliability).

Threats to the validity of a study

Internal validity: integrity of findings

- Threatened by **confounding variables**: additional, often unmeasured variables that are related to both the predictor and the outcome

External validity: generalisability of findings

- Threatened by **artefacts**: factors that are unique to the special situation in which you carry out your research; they could render your findings applicable only to the artificial conditions of the study (e.g., findings apply only to performance on artificial lab-based tasks, not real-world behaviour).